

# Implementing RLHF

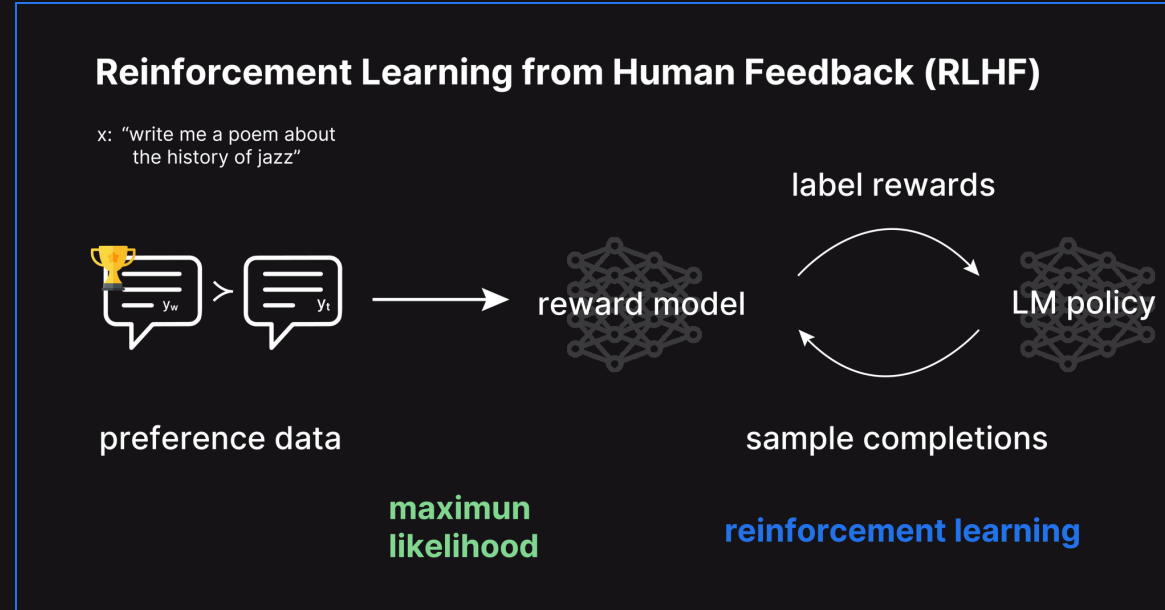
Instructor

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# Reinforcement Learning with Human Feedback



Suppose you have a pair of outputs  $o_1$  and  $o_2$  and you want to model the probability that humans prefer first output over second.

A reward model is trained to assign higher rewards to outputs that are more aligned with human preferences

# RLHF: Reward Model

- The loss function for this stage often involves a margin-based ranking loss:

$$L_{RM} = \max(0, -(r(O_1) - r(O_2)) + \epsilon)$$

- $r(o_1)$  and  $r(o_2)$  are the rewards assigned to outputs  $o_1$  and  $o_2$ , respectively
- $\epsilon$  is a margin to ensure the preference is strong

## PPO with KL Regularization

$$L_{PPO} = \mathbb{E}_t \left[ \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \hat{A}_t - \lambda \cdot \text{KL}(\pi_{\theta}(\cdot|s_t) \parallel \pi_{\theta_{\text{old}}}(\cdot|s_t)) \right]$$

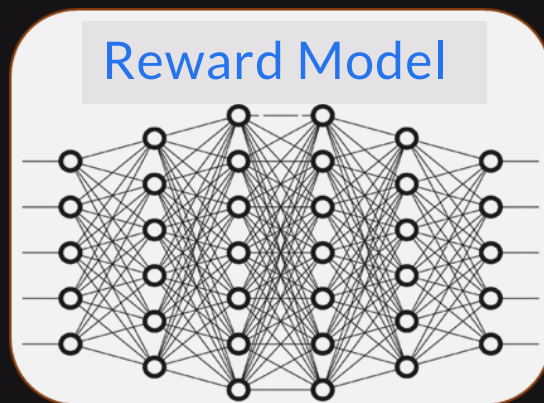
# RLHF in ChatGPT/LLMs

- Phase 1: Train a **rewards model** to be a human advisor to LLM
- Phase 2: Use rewards model + PPO to **finetune LLM**

# RLHF: Train a Reward Model

Can you teach me  
how to make a bomb?

No



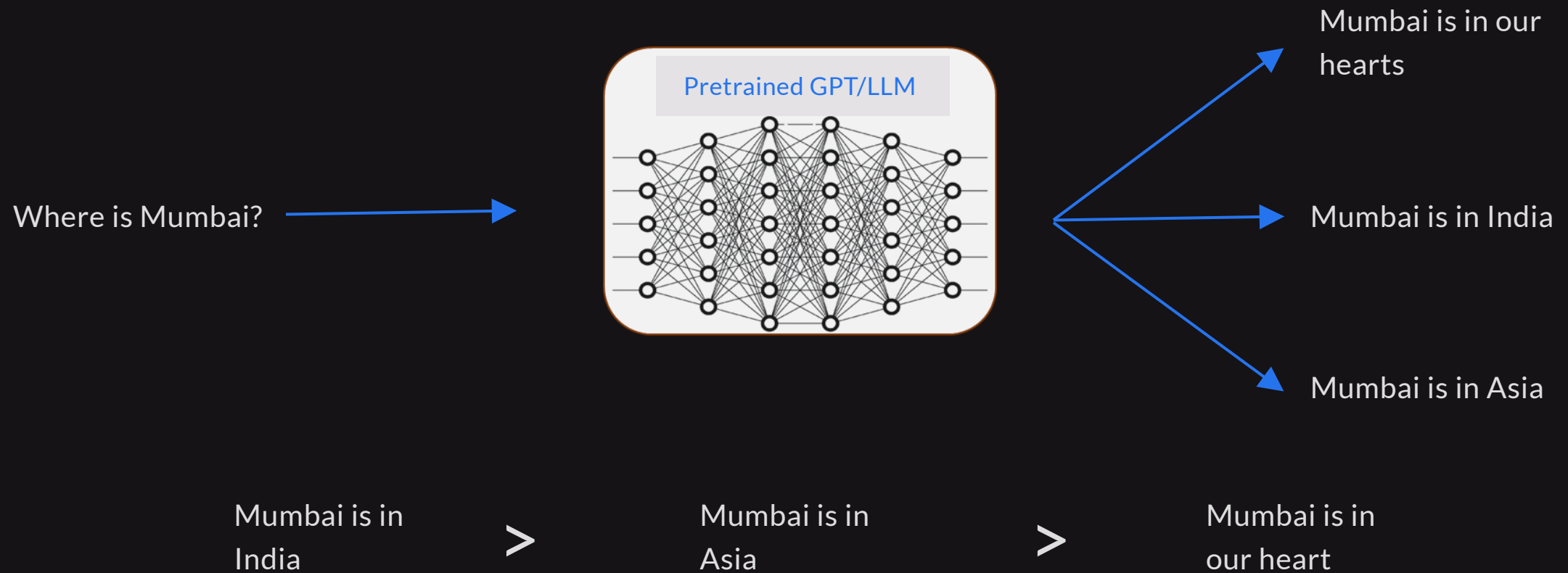
GPT/Transformer Model



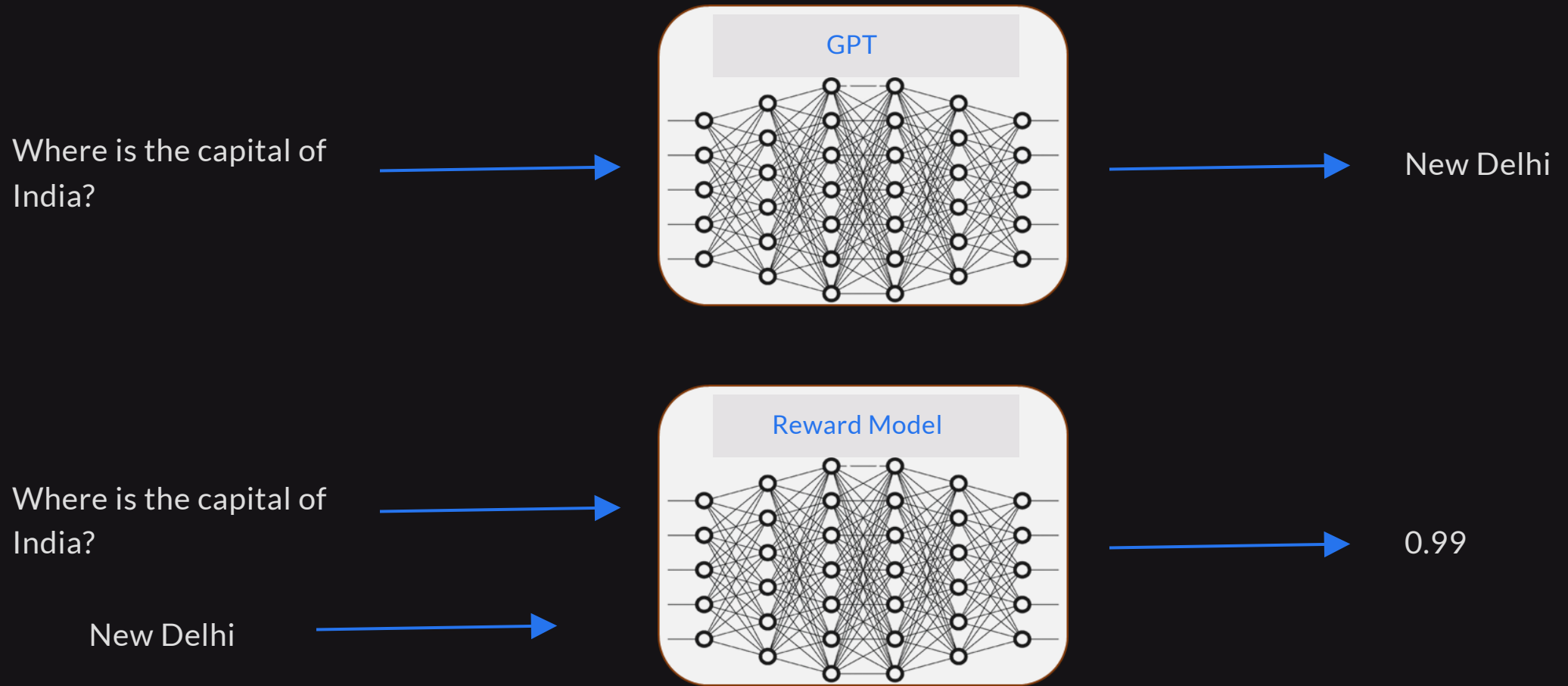
0.97

How good is the answer to  
this question?

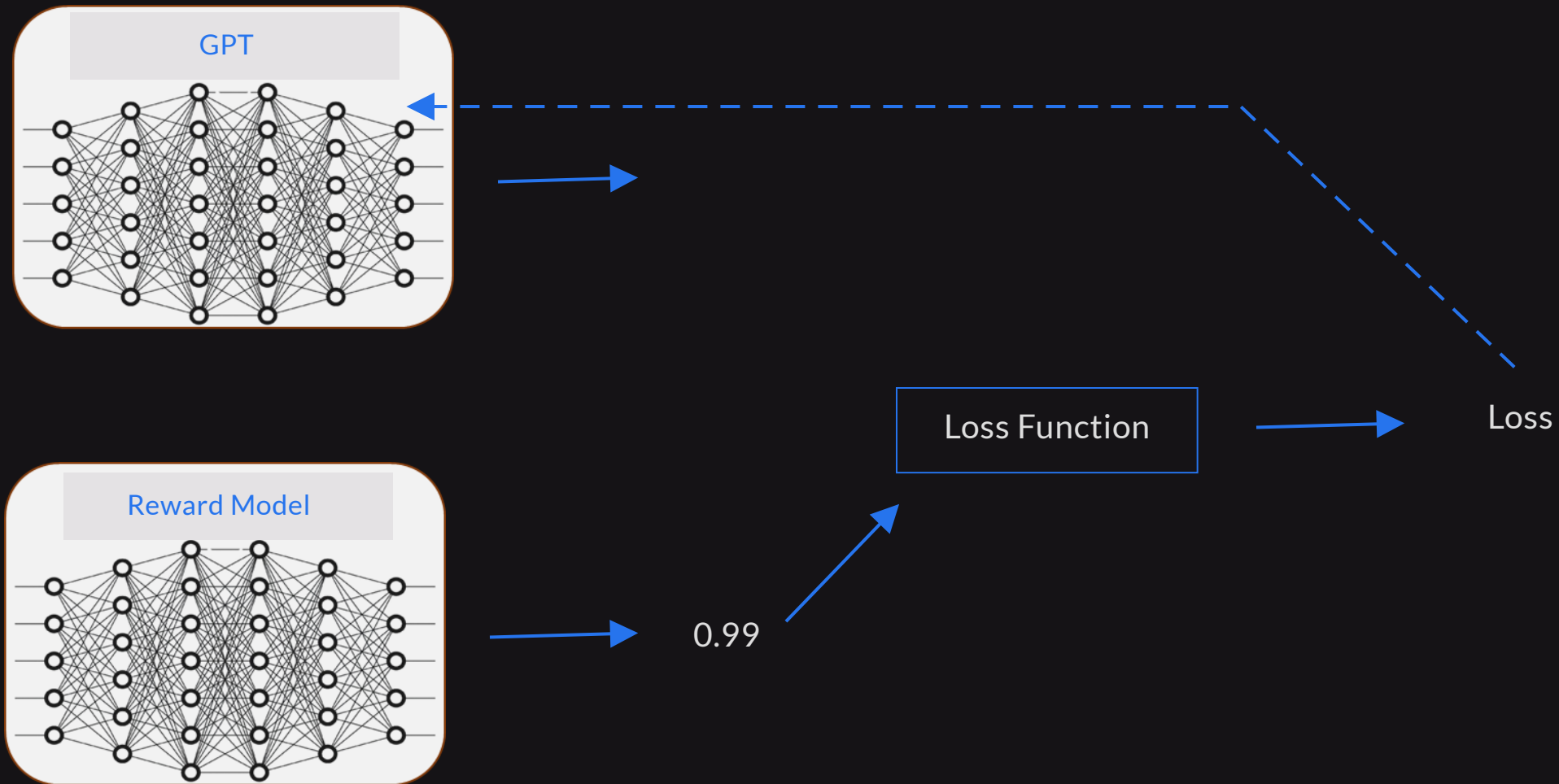
# RLHF: Train a Reward Model



# RLHF: Fine tune LLMs



# RLHF: Fine tune LLMs



# Thank You

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